

AVSD: Adaptive-View Self-Distillation by Balancing Consensus and Teacher-Specific Privileged Signals

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Abstract

Self-distillation enables language models to learn on-policy from their own trajectories by using the same model as both student and teacher, with the teacher being conditioned on privileged information unavailable to the student. Such information can come in different types or views, such as solutions, demonstrations, feedback, or final answers. This setup provides dense token-level feedback without relying on a separate external model, but creates a fundamental *asymmetry*: the teacher may rely on view-specific information that the student cannot access at inference time. **Moreover, the best type of privileged information is often task-dependent, making it difficult to choose a single teacher view.** In this work, we address both these challenges jointly by introducing AVSD (**A**daptive-**V**iew **S**elf-**D**istillation), **a novel method of self-distillation with multiple privileged-information views, which reconstructs token-level supervision by separating stable cross-view consensus from view-specific residual signals.** AVSD identifies the consensus signal shared across views, which provides a reliable update direction, and then selectively adds the view-specific residual signal to adjust the update magnitude when it both aligns with the consensus direction and remains proportionate to the consensus signal. Experiments on math competition benchmarks (AIME24, AIME25, and HMMT25) show that AVSD consistently outperforms both single-view self-distillation baselines and GRPO, achieving average Avg@8 gains of 3.1% and 2.2% over the strongest baselines on Qwen3-8B and Qwen3-4B, respectively. Moreover, on code-generation benchmarks (Codeforces, LiveCodeBench v6) using Qwen3-8B, AVSD outperforms the single-view self-distillation baseline by 2.4% on average.¹

1 Introduction

Training large language models (LLMs) to solve difficult reasoning tasks, such as competition-level math problems requires converting available training supervision into useful learning signals. Reinforcement learning with verifiable rewards (RLVR) [Shao et al., 2024, DeepSeek-AI, 2025] has become the dominant paradigm for LLM post-training in verifiable tasks such as math and code, but its supervision is typically sparse and outcome level, leading to expensive sampling and limited learning signal when the model fails to produce a successful final solution [Tao et al., 2025]. **Distillation offers a complementary source of dense token-level guidance [Hinton et al., 2015],** but standard offline distillation suffers from train-test mismatch because the student is trained on trajectories outside its own distribution [Xu et al., 2024, Agarwal et al., 2024]. On-policy distillation addresses this off-policy issue by training on student-generated rollouts while using teacher probabilities to provide local learning signals [Agarwal et al., 2024, Gu et al., 2024]. However, these methods still rely on access to a more capable teacher model, **motivating self-distillation methods,** which remove the need for an external teacher by employing the student model as its own teacher, supplying a learning signal

¹Code: <https://github.com/duykhuongnguyen/AVSD>.

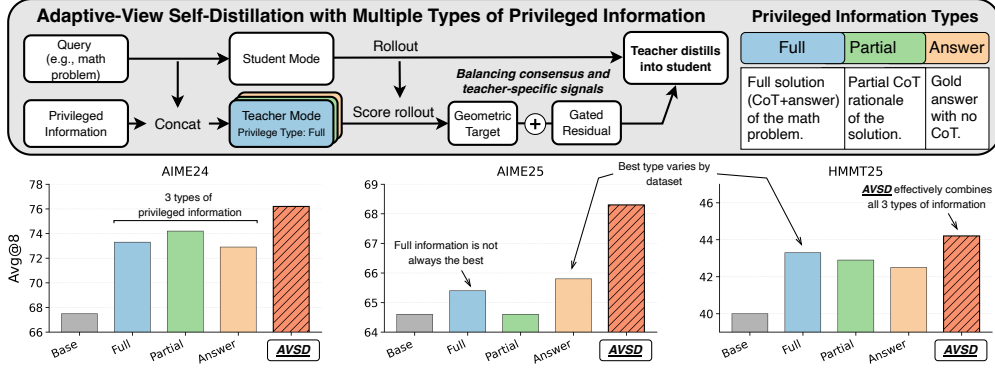


Figure 1: Self-distillation leverages access to privileged ground truth information as a teacher to distill into the student. However, performance across datasets on Qwen3-4B varies substantially depending on the type of privileged ground truth information the teacher has access to (full solution, partial reasoning, or only final answer). We show that no single type is uniformly optimal, and that AVSD consistently achieves the best performance across datasets by effectively combining multiple types of privileged information.

by giving the teacher privileged information that the student cannot access. For example, the student model might learn from a teacher with access to different augmented context views, such as solutions, demonstrations, or feedback [Zhao et al., 2026, Shenfeld et al., 2026].

Current self-distillation methods share a strong and often implicit assumption: once a privileged context type is chosen, it remains fixed throughout training [Shenfeld et al., 2026, Penaloza et al., 2026]. However, in practice, a variety of privileged information types are possible, with no clear way of determining *a priori* which will be optimal. For example, for a math problem, one may have access to a verified full solution, a final answer, a partial rationale, or a concise hint (see Fig. 1 bottom-center, showing differences when using full vs partial vs answer only privileged information). **Each type comes with its own limitations**, e.g., providing a full solution with chain of thought may lock the teacher into a preexisting thought pattern that does not align with the student’s, while providing only the final answer may be more flexible but less informative. These different types of information can result in meaningful performance differences: Penaloza et al. [2026] show that different forms of privileged information vary in information density, task utility, and **induced student-teacher distribution shift**, and that the best choice depends on the task and model regime. Yang et al. [2026] show that distribution matching under information asymmetry contains an irreducible mutual-information gap, so a student trained to imitate a privileged teacher is pressured to encode view-specific correlations it cannot observe at test time. Kim et al. [2026] show that the most informative privileged form discards important signals that help models identify reasoning mistakes, and degrading out-of-domain mathematical reasoning, even when training traces are correct. These observations suggest that choosing a single privileged view has important consequences for performance and generalization, and that no single view is guaranteed to provide the best learning signals across tasks. This raises the central question of our work: **can we leverage multiple privileged views simultaneously to construct better token-level on-policy learning signals for self-distillation than any single privileged teacher?**

In this work, we propose **Adaptive-View Self-Distillation (AVSD)**, an on-policy self-distillation framework that uses multiple types of privileged information rather than committing to a single teacher context. The key intuition is: (1) if different privileged views induce similar token-level updates, then the signal is likely stable and task-relevant. (2) In contrast, if a token is strongly favored by only one view, that signal may still be useful because it may capture complementary information not emphasized by other views, but it is riskier because it may depend on information that the student cannot access at inference time. AVSD formalizes this intuition by separating the teacher signal of multiple views into a shared part and an extra view-specific part. We refer to the shared part as the *consensus* signal: it captures tokens that are consistently supported across privileged views and provides a reliable update direction. We refer to the extra view-specific part as the *residual*: it captures additional support that appears when one or a few views strongly favor a token beyond what is shared by all views. To expose these two parts, AVSD uses two pooled targets: the geometric consensus

target and the arithmetic marginal target. The *geometric consensus target* emphasizes tokens that are supported across views, while the *arithmetic marginal target* preserves tokens that receive strong support from at least one view. Rather than distilling from either target directly, AVSD starts from the consensus signal and uses a gate to add the residual only when the different views agree on the promote-or-suppress direction, and the residual magnitude remains proportionate to the consensus magnitude. This allows AVSD to benefit from complementary information in different privileged views while preventing any single view from dominating the consensus token-level learning signal.

We validate AVSD on math and code reasoning benchmarks. Empirically, using three automatically-constructed privileged views for each math problem (full solution, partial solution, and final answer), we show that AVSD outperforms the standard single-view self-distillation baseline [Zhao et al., 2026, Shenfeld et al., 2026] and GRPO [Shao et al., 2024] on math competition benchmarks (AIME24 [Zhang and Math-AI, 2024], AIME25 [Zhang and Math-AI, 2025], and HMMT25 [Balunovic et al., 2025]). On Qwen3-4B [Yang et al., 2025], AVSD improves Avg@8 by 5.5% over the base model and 2.2% over the strongest baseline. The gains are consistent at larger scale: on Qwen3-8B, AVSD achieves the best average score, improving over the strongest baseline by 3.1%. We also observe similar improvements on DeepSeek-R1-Distill-Qwen-7B [DeepSeek-AI, 2025], where AVSD improves over the self-distillation baseline by 2.3% on average. Additionally, on coding benchmarks (Codeforces [Penedo et al., 2025], LiveCodeBench v6 [Jain et al., 2025]), AVSD improves the single-view self-distillation baseline by 2.4% on Qwen3-8B. Ablations show that consensus-only and arithmetic-only variants both underperform AVSD, and our token-level analysis further confirms that AVSD provides a better learning signal by jointly preserving cross-view agreement and selectively adding gated residual support.

2 AVSD: Adaptive-View Self-Distillation

In this section, we present AVSD (Adaptive-View Self-Distillation). In Section 2.1, we first formulate on-policy self-distillation through the token-level reverse-KL advantage, then extend it to the multi-view setting where each privileged view induces a different teacher distribution.

At a high level, AVSD builds a teacher signal by separating consensus from view-specific signal. If several views agree that a token should be promoted or suppressed, this provides a reliable anchor for the student update. In Section 2.2, we capture this shared signal with a *geometric consensus target*, which emphasizes tokens supported across views. In contrast, a token favored by only one or a few views may capture complementary task-relevant information, but may also reflect privileged artifacts. We expose this extra signal with an *arithmetic marginal target*, and define the *residual* as its difference from the geometric consensus target. Finally, in Section 2.3, we propose a gate to reconstruct the teacher signal by selectively adding this residual when the views agree on the update direction and the residual magnitude remains proportionate to the consensus magnitude.

2.1 Problem Setup

Let $\mathcal{D}$ be a training set. For notation simplicity, we write $(x, r) \sim \mathcal{D}$ for a generic example, where x is a prompt and r is the training-time privileged information. The student observes only x , while the teacher conditions on r during training. During training, we sample an on-policy rollout

$$y = (y_1, \dots, y_T) \sim P_\theta^S(\cdot | x).$$

We denote the prefix at position t as $h_t = (x, y_{<t})$. The student next-token distribution over vocabulary $\mathcal{V}$ is

$$p_t(v) := P_\theta^S(Y_t = v | h_t), \quad v \in \mathcal{V},$$

where v denotes a token type in the vocabulary. Standard single-view self-distillation chooses one view r and uses the same model conditioned on that view as the teacher. At the same prefix, denote

$$q_t(v) := \text{sg}[P_\theta^T(Y_t = v | h_t, r)], \quad v \in \mathcal{V},$$

where P_θ^T denotes the same model evaluated in the privileged teacher context and $\text{sg}[\cdot]$ is the stop gradient operator. The single-view reverse-KL [Gu et al., 2024, Agarwal et al., 2024] objective is

$$\mathcal{L}_{\text{sv}}(\theta) = \mathbb{E}_{(x,r) \sim \mathcal{D}} \mathbb{E}_{y \sim P_\theta^S(\cdot | x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} D_{\text{KL}}(p_t \| q_t) \right].$$

The negative local gradient of the reverse-KL term can be written in **policy-gradient form** as

$$-\nabla_{\theta} D_{\text{KL}}(p_t \parallel q_t) = \mathbb{E}_{v \sim p_t} [A_t(v) \nabla_{\theta} \log p_t(v)],$$

where

$$A_t(v) := \log q_t(v) - \log p_t(v). \quad (1)$$

$A_t(v)$ in Eq. (1) is the token-level distillation advantage. If $A_t(v) > 0$, the teacher assigns token type v higher probability than the student, so gradient descent on the reverse-KL objective promotes v ; if $A_t(v) < 0$, the update suppresses v . The full derivation of this policy-gradient form from the reverse-KL objective is given in Appendix B.1. We use $A_t(v)$ to describe the sign and relative magnitude of the token-level updates. In the single-view setting, this corresponds to choosing one privileged information type throughout training, such as always conditioning the teacher on a full solution in Fig. 1.

In the multi-view setting, we construct **M privileged views** $\mathcal{V}(r) = \{r^{(1)}, \dots, r^{(M)}\}$, where $r^{(m)} = T_m(r)$ is the transformation that preserves task-relevant information while changing which privileged view is revealed to the teacher. For example, for a math problem, views may be a full solution, final answer, or partial rationale. Each view induces a teacher distribution at the same student prefix, which shows how different privileged information types would promote or suppress the same generated token (examples in Fig. 2 and Fig. 4):

$$q_t^{(m)}(v) := \text{sg} \left[P_{\theta}^T(Y_t = v \mid h_t, r^{(m)}) \right], \quad m = 1, \dots, M,$$

The corresponding per-view distillation advantage is $\Delta_t^{(m)}(v) := \log q_t^{(m)}(v) - \log p_t(v)$. A key question then becomes: **Given a teacher family $\{q_t^{(m)}\}_{m=1}^M$, how should we weigh and combine distributions from teachers to determine the token-level signal that the student receives?**

2.2 Consensus and Residual Signal

Different privileged views can agree or disagree at the same student prefix on the *signed token-level update* induced by each teacher view. For a token v , the teacher family provides a **stable signal** when the per-view advantages $\Delta_t^{(m)}(v)$ have consistent signs and comparable magnitudes: multiple views either agree that v should be promoted or agree that it should be suppressed. This *consensus* is informative because all teachers are conditioned on the same student-visible prefix h_t and differ only in what training-time privileged information they have access to. Therefore, an update that persists across several privileged views is less likely to depend on a particular view-specific artifact and more likely to reflect a *consensus* signal that is compatible with the information available from h_t . At the same time, **not all useful information needs to be shared by every view**: a token may receive strong support from only a subset of views because those views capture reasoning paths complementary to the consensus signal (Fig. 2 Token 2). However, the same subset-specific support may also reflect privileged artifacts that the student cannot access at test time (Fig. 2 Token 1). To resolve this, we use **two pooled targets**: one target captures tokens that are consistently supported across views, while the other captures tokens that are strongly supported by at least one view. **Their gap defines the residual signal**, which AVSD later gates before adding it to the consensus update.

Geometric Consensus Target. The first pooled target captures intersection support: **a token receives large probability only if it is supported across the teacher family**. We define the unnormalized geometric score and its normalized distribution as

$$\tilde{q}_t^G(v) := \left(\prod_{m=1}^M q_t^{(m)}(v) \right)^{1/M} = \exp \left(\frac{1}{M} \sum_{m=1}^M \log q_t^{(m)}(v) \right), \quad q_t^G(v) = \frac{\tilde{q}_t^G(v)}{\sum_{u \in \mathcal{V}} \tilde{q}_t^G(u)}.$$

We call q_t^G the geometric consensus target as it is the normalized geometric mean of the view-conditioned teachers. The geometric mean downweights tokens that are assigned low probability by any view, and therefore emphasizes tokens that are jointly compatible with the teacher family. Thus, this target can be conservative (Fig. 2 top-center). Equivalently, q_t^G is the distribution that minimizes average reverse KL to the teacher. The full derivation is in Appendix B.

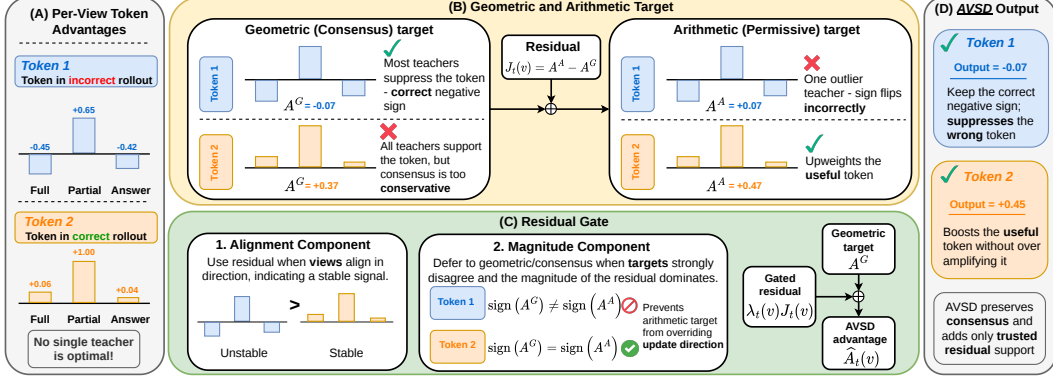


Figure 2: (A) Bar graphs show teacher advantages on student tokens. (B) The geometric target demands consensus between teachers (conservative), while the arithmetic target allows individual teachers to contribute more strongly (permissive). The residual is their difference. (C) A two-component gate controls the residual: the alignment component suppresses unstable signals when views conflict, and the magnitude component prevents the residual from flipping the update direction established by consensus. (D) AVSD combines both: it preserves the consensus direction on risky tokens and boosts useful view-specific signals on tokens with high cross-view alignment.

For token-level updates, we use the unnormalized geometric score, since the normalization contributes only a token-independent constant

$$A_t^G(v) := \log \tilde{q}_t^G(v) - \log p_t(v) = \frac{1}{M} \sum_{m=1}^M \Delta_t^{(m)}(v).$$

Thus, $A_t^G(v)$ is the average signed update induced by the teacher family.

Arithmetic Marginal Target. The second pooled target captures union support: a token can receive high probability if it is strongly supported by any view (Fig. 2 top-center). We define

$$q_t^A(v) := \frac{1}{M} \sum_{m=1}^M q_t^{(m)}(v).$$

This is the arithmetic mean of the view-conditioned teachers. Equivalently, if a privileged view index is sampled uniformly at random, q_t^A is the marginal next-token distribution obtained after integrating out the view. It is also the solution to the average forward-KL aggregation problem that finds the single pooled target that minimizes the average forward-KL from the view-conditioned teachers (see Appendix B.2 for details). The arithmetic advantage is

$$A_t^A(v) := \log q_t^A(v) - \log p_t(v).$$

The arithmetic target preserves complementary support that appears in only one or a few views, which can be valuable when different views reveal different valid reasoning paths. However, this permissiveness also makes it sensitive to view-specific artifacts. For example, a full-solution view may strongly promote a token because it has access to a reference derivation, while an answer-only view provides no support for the same token. Therefore, directly distilling from the arithmetic target would lead to an update based on a noisy signal.

Cross-View Residual. Because q_t^G is conservative and q_t^A is permissive, their gap isolates the support retained by the permissive marginal but discarded by the strict consensus:

$$A_t^A(v) = A_t^G(v) + J_t(v), \quad J_t(v) := \log q_t^A(v) - \log \tilde{q}_t^G(v) \geq 0.$$

We call $J_t(v)$ the *cross-view residual* (Fig. 2 top-center). It is non-negative by the arithmetic-geometric mean inequality, since $q_t^A(v) \geq \tilde{q}_t^G(v)$ for every token v . Intuitively, $J_t(v)$ measures how much probability mass is present in the arithmetic marginal but absent from strict geometric consensus. A large residual indicates that q_t^A preserves support from only a subset of teachers, while

a zero residual indicates exact token-level consensus across all teachers. The residual alone does not determine whether the extra support is reliable, because the same large residual can arise from useful complementary evidence or view-specific artifacts. Therefore, we use the consensus as the update direction and use the residual to adjust the magnitude of the update.

2.3 Gated Reconstruction of Token-Level Supervision via Consensus and Residual Signal

Residual Gate. The decomposition above suggests a simple principle: always keep view consensus, and add the cross-view residual only when it is reliable. We define the reconstructed advantage:

$$\hat{A}_t(v) := A_t^G(v) + \lambda_t(v)J_t(v), \quad \lambda_t(v) \in [0, 1].$$

The gate $\lambda_t(v)$ is designed around two requirements, illustrated by Token 1 and Token 2 in Fig. 2. For Token 1, one view strongly promotes the token, but the other views suppress it. In this case, the arithmetic target would incorrectly flip the update direction, so the gate should defer to the consensus signal. For Token 2, all views promote the token, but one view provides much stronger evidence than the others. In this case, the residual can be useful, because it strengthens an update direction that the views already agree on. However, it should still be bounded so that the extra view-specific signal does not dominate the consensus. These two cases motivate two components of the gate described below.

First, the residual should not be trusted when teacher views heavily contradict each other in terms of update direction. For example, if some views promote a token while others suppress it, the residual is unreliable (Fig. 2 Alignment Component). We define the alignment component by the ratio between the magnitude of the averaged signed advantage and the average magnitude of the per-view advantages:

$$C_t(v) := \frac{|A_t^G(v)|}{\frac{1}{M} \sum_{m=1}^M |\Delta_t^{(m)}(v)| + \epsilon} = \frac{\frac{1}{M} |\sum_{m=1}^M \Delta_t^{(m)}(v)|}{\frac{1}{M} \sum_{m=1}^M |\Delta_t^{(m)}(v)| + \epsilon} \in [0, 1],$$

$C_t(v)$ is high when the per-view advantages have the same sign or when unsupported views are nearly neutral, and low when positive and negative view-specific updates cancel. This allows AVSD to distinguish complementary support from conflict-driven support.

Second, the residual should be proportionate to the consensus signal (Fig. 2 Magnitude Component). If $J_t(v)$ is much larger than $|A_t^G(v)|$, then the arithmetic residual can dominate or even reverse the consensus update. For example, if $A_t^G(v) < 0$ but $J_t(v) \gg |A_t^G(v)|$, the arithmetic target would promote a token that the teacher family suppresses on average. To prevent this, we define:

$$R_t(v) = \frac{|A_t^G(v)|}{|A_t^G(v)| + J_t(v) + \epsilon}.$$

The final gate is

$$\lambda_t(v) = C_t(v)R_t(v) = C_t(v) \frac{|A_t^G(v)|}{|A_t^G(v)| + J_t(v) + \epsilon}, \quad (2)$$

This gate acts as an adaptive regularizer on the residual signal. It admits residual support when the *consensus is coherent* and the *residual is proportionate*, but suppresses it when the views conflict, when the consensus is weak, or when the residual is too large. Moreover, expanding the gate in Eq. (2) gives $\lambda_t(v)J_t(v) \leq |A_t^G(v)|$, since $C_t(v) \leq 1$ and $J_t(v) \leq |A_t^G(v)| + J_t(v) + \epsilon$ (the full derivation is provided in Appendix B.3); the residual can adjust the update magnitude but cannot reverse the update direction induced by the consensus. Our gate also connects to the failure mode identified by Yang et al. [2026]: directly matching a teacher conditioned on information unavailable to the student can induce an irreducible mutual-information gap and privileged-information leakage. AVSD addresses this risk in the multi-view setting by preventing subset-supported residuals from freely determining the update direction. Fig. 2 illustrates how the gate addresses failure modes of the two pooled targets. For Token 1, one view promotes the token while the others suppress it, so AVSD keeps the consensus direction and prevents the residual from flipping the update. For Token 2, the views agree on the positive direction, so the residual can strengthen the update.

Reconstructed Target. The reconstructed target corresponding to $\hat{A}_t$ is obtained by reweighting the geometric consensus target with the gated residual and renormalizing:

$$q_t^*(v) = \frac{\tilde{q}_t^G(v) \exp(\lambda_t(v)J_t(v))}{\sum_{u \in \mathcal{V}} \tilde{q}_t^G(u) \exp(\lambda_t(u)J_t(u))}.$$

This interpolates tokenwise between two targets. If $\lambda_t(v) = 0$, the method reduces to the geometric consensus target. If $\lambda_t(v) = 1$ for all v , then the method recovers the arithmetic marginal target.

Training Objective. We train the student by on-policy reverse-KL distillation to the reconstructed target. The distillation loss is

$$\mathcal{L}_{\text{AVSD}}(\theta) = \mathbb{E}_{(x,r) \sim \mathcal{D}} \mathbb{E}_{y \sim P_{\theta}^S(\cdot|x)} \left[\sum_{t=1}^{|y|} D_{\text{KL}}(p_{\theta}(\cdot | h_t) \parallel \text{sg}[q_t^*(\cdot)]) \right], \quad (3)$$

Gradients flow only through the student distribution $p_{\theta}(\cdot | h_t)$, while the teacher distributions and the reconstructed target are treated as fixed supervision at the current prefix. Equivalently, using the log-derivative form of the reverse-KL objective, the sampled-token update is the reconstructed advantage $\hat{A}_t(y_t) = A_t^G(y_t) + \lambda_t(y_t) J_t(y_t)$. Compared with standard single-view self-distillation, AVSD only replaces a single teacher-context evaluation with M view-conditioned teacher evaluations at the same student prefix. It does not require additional student rollouts: all views share the same generated trajectory and can be evaluated in parallel or batched. Since on-policy rollout generation is sequential and typically dominates the training cost, while the additional view-conditioned teacher evaluations are prefill-only forward passes, AVSD only adds modest overhead in practice (details in Table 2). We provide the full training algorithm in Appendix B.4.

3 Experiments

In this section, we evaluate AVSD on math and code reasoning benchmarks across three base models. We first present the experimental setup, including datasets, baselines, and evaluation, and then present the main empirical results.

3.1 Experimental Setup

Models. We evaluate across multiple model families and scales, including Qwen3-4B and Qwen3-8B [Yang et al., 2025], and DeepSeek-R1-Distill-Qwen-7B [DeepSeek-AI, 2025].

Datasets. We use the following math and code datasets for training and evaluation:

- **Math datasets:** Following Zhao et al. [2026], we use the mathematical reasoning subset of OpenThoughts [Guha et al., 2026] as the training data. For the multi-view setup, we use three views that are directly available or easily derived from the dataset: the full solution, the final answer, and a partial solution obtained by retaining the initial reasoning steps of the full solution. We evaluate on competition-level mathematics benchmarks including AIME 2024 [Zhang and Math-AI, 2024], AIME 2025 [Zhang and Math-AI, 2025], and HMMT 2025 [Balunovic et al., 2025].
- **Code datasets:** We train the models on Codeforces [Penedo et al., 2025], sampling 5K examples from the Python subset. For each problem, we construct three privileged views: a reference implementation, an algorithmic hint derived from the reference solution, and execution feedback obtained by running the student rollout on available test cases, which includes the generated program and the verifier’s pass/fail results. For evaluation, we hold out 100 Codeforces problems as an in-domain test set and further evaluate on LiveCodeBench v6 [Jain et al., 2025].

The examples of math and code views are provided in Appendix E. Similar to [He et al., 2026], we report Avg@8 throughout the experiments. We provide the detailed training and evaluation hyperparameters in Appendix C.

Baselines. We compare against three methods trained on the same dataset: (1) SFT, standard supervised fine-tuning, which can be seen as off-policy distillation from a static dataset of the reasoning traces, (2) GRPO [Shao et al., 2024], group relative policy optimization with binary outcome rewards verified against ground-truth answers, (3) OPSD, the single-view on-policy self-distillation baseline [Zhao et al., 2026, Shenfeld et al., 2026]. For this baseline, we use the same

Table 1: Avg@8 accuracy on math and code benchmarks across three base models. The best result within each model block is in **bold**, and the second best is underlined.

Method	Math			Code		Avg.
	AIME24	AIME25	HMMT25	CF	LCB	
<i>Qwen3-4B</i>	67.5	64.6	40.0	59.8	43.2	55.0
+ SFT	56.3	52.1	29.6	59.1	40.7	47.6
+ GRPO	70.6	<u>65.7</u>	41.8	61.2	43.9	56.6
+ OPSD	<u>73.3</u>	65.4	<u>43.3</u>	<u>63.6</u>	<u>45.2</u>	<u>58.2</u>
+ AVSD (Ours)	76.2	68.3	44.2	65.6	45.4	59.9
<i>Qwen3-8B</i>	71.2	65.4	41.2	60.9	47.6	57.3
+ SFT	70.8	63.5	41.5	61.2	43.2	56.0
+ GRPO	72.5	<u>67.1</u>	<u>43.0</u>	62.8	49.9	59.1
+ OPSD	<u>74.2</u>	<u>65.8</u>	<u>42.5</u>	<u>63.2</u>	50.3	<u>59.2</u>
+ AVSD (Ours)	75.4	69.6	47.1	65.8	52.4	62.1
<i>DeepSeek-R1-Distill-Qwen-7B</i>	48.3	37.9	21.7	44.8	35.7	37.7
+ SFT	48.5	36.2	20.5	41.6	35.2	36.4
+ GRPO	51.2	37.8	<u>23.1</u>	<u>46.2</u>	<u>38.0</u>	<u>39.3</u>
+ OPSD	<u>53.3</u>	37.5	22.6	43.6	37.6	38.9
+ AVSD (Ours)	55.8	39.2	25.4	47.4	39.1	41.4

reverse-KL distillation objective as AVSD, but condition the teacher on a single privileged view: the full solution for math and the reference implementation for code, following the original OPSD setting [Zhao et al., 2026].

3.2 Results: AVSD Improves Math and Code Reasoning over GRPO and OPSD

Table 1 shows that AVSD consistently improves both math and code reasoning across model families. On the math benchmarks, AVSD achieves the best average Avg@8 for all three base models. For Qwen3-4B, AVSD improves over the base model by 5.5% and outperforms the strongest baseline, OPSD, by 2.2% on average. The gains are consistent at larger scale: on Qwen3-8B, AVSD improves over the base model by 4.8% and over the strongest baseline, GRPO, by 3.1%. We observe similar improvements on DeepSeek-R1-Distill-Qwen-7B, where AVSD improves over the strongest self-distillation baseline by 2.3% on average.

The benefits also transfer to code generation. On Qwen3-8B, AVSD improves the average code Avg@8 on Codeforces and LiveCodeBench over the base model and OPSD by 4.9% and 2.4%, respectively. Similar trends hold for Qwen3-4B and DeepSeek-R1-Distill-Qwen-7B, where AVSD achieves the best code average among the compared methods. Overall, these results demonstrate that AVSD provides consistent gains over GRPO and single-view on-policy self-distillation baselines across both math and code benchmarks.

4 Analysis

In this section, we analyze AVSD and examine why it improves over single-view self-distillation. We first ablate the consensus and arithmetic targets to isolate the role of gated reconstruction, and then examine token-level credit allocation to test whether AVSD provides more reliable learning signals on student-generated rollouts. Finally, we study how performance scales with the number of privileged views, testing whether AVSD benefits from richer and more diverse teacher families.

4.1 Ablating Consensus and Arithmetic Targets

We evaluate the performance of AVSD when using either the consensus target or the arithmetic marginal target alone. The consensus-only variant uses A_t^G as the learning signal and discards the arithmetic-over-consensus residual. The arithmetic-only variant distills directly from q_t^A . This comparison isolates the role of our gated reconstruction. Fig. 3 shows that neither standalone aggregation is uniformly optimal: the consensus target performs strongly on AIME24 and AIME25 but

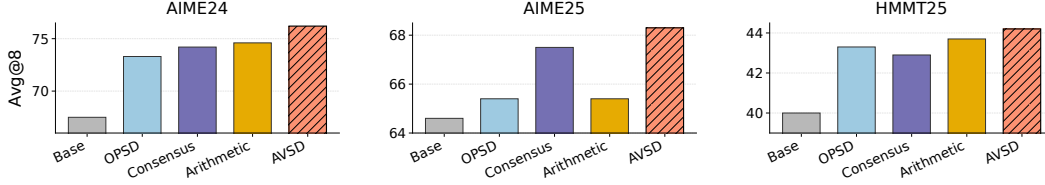


Figure 3: Comparison of single-view OPSD, consensus-only distillation, arithmetic-only distillation, and the full AVSD on Qwen3-4B. AVSD achieves the best Avg@8 across all datasets.

is slightly worse than OPSD on HMMT25, while the arithmetic target improves on AIME24 and HMMT25 but gives limited gains on AIME25. In contrast, AVSD achieves the best performance across all benchmarks and improves over both standalone multi-view targets. Averaged across the three benchmarks, AVSD outperforms the consensus-only and arithmetic-only variants by 1.4% and 1.7%, respectively. This demonstrates that combining the consensus signal with a selectively gated residual is necessary: the consensus target alone can be overly conservative, while the arithmetic target alone can overuse view-specific support. These results support our central claim that reliable multi-view self-distillation requires reconstructing a token-level target from both stable cross-view agreement and carefully filtered residual information.

4.2 Token-Level Credit Allocation

We further analyze whether the token-level learning signal produced by AVSD assigns credit in the desired direction. The sampled-token advantage determines whether the probability of a generated token is increased or decreased. For an incorrect rollout, generated tokens should generally receive negative credit, indicating that the method suppresses erroneous solutions. To evaluate this behavior, we collect student rollouts whose final answers are incorrect and compute the sampled-token advantage at every generated token. For OPSD and AVSD, we select the top 20 tokens with the largest absolute advantage magnitude, corresponding to the tokens where the distillation signal has the strongest effect on training, and measure the fraction of these tokens whose credit has the incorrect (positive) sign. For GRPO, which does not provide dense token-level supervision, we instead report the fraction of cases with no learning signal (e.g., while rollouts should have either positive or negative signs depending on correctness, by construction, most GRPO methods give no signal when no sampled rollout is correct). The heatmap in Fig. 4 visualizes this behavior at the token level for the top 5 tokens in one rollout: each column is a generated token from the incorrect rollout, and each cell shows whether a method increases or decreases that token’s probability. Since the rollout is incorrect, the desired behavior is to assign negative credit to the high-impact tokens, i.e., the generated tokens with the largest absolute advantage magnitudes and therefore the strongest effect on the training loss. Single-view baselines produce mixed signs across tokens, suggesting that their feedback depends strongly on the chosen privileged view, whereas AVSD assigns more consistently negative credit to the key tokens while avoiding the near-zero signal of GRPO. Quantitatively, GRPO has no learning signal in 29.6% of all examples, OPSD assigns incorrect-sign credit to 12.3% of high-impact tokens, whereas AVSD reduces this rate to only 2.9% (see Fig. 8, Appendix D). These results show that

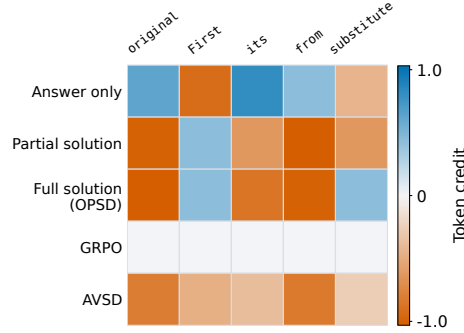


Figure 4: Example from a single *incorrect rollout*: we show the top-5 generated tokens with largest absolute token-credit magnitude from that rollout. Positive values indicate that the token is encouraged, while negative values indicate that the token is discouraged. For GRPO, the sampled group for the same problem contains no correct rollout, so the outcome-level reward provides no learning signal. **Single-view self-distillation teachers assign inconsistent or misleading signs across tokens, whereas AVSD provides a more coherent credit signal on key tokens.** Full problem and generation are in Appendix D.

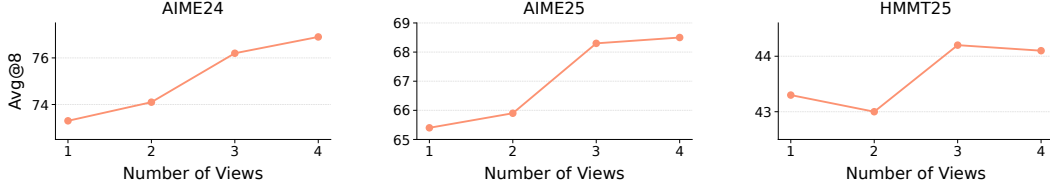


Figure 5: Scaling the number of privileged math views for AVSD on Qwen3-4B.

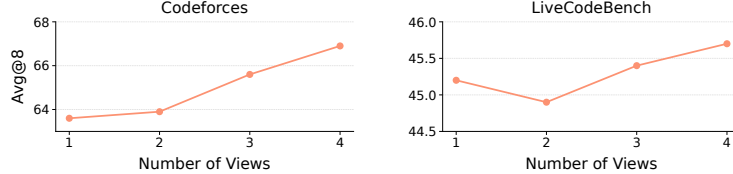


Figure 6: Scaling the number of privileged code views for AVSD on Qwen3-4B. Final performance generally improves as more complementary privileged views are added.

AVSD provides a substantially more reliable token-level learning signal than both sparse-reward GRPO and single-view self-distillation.

4.3 Scaling the Number of Views

We study how performance changes as we increase the number of privileged views used by AVSD. For math, we start from a single full-solution view and incrementally add a partial-solution view, the final-answer view used in our main experiments, and a fourth view that conditions the teacher on the model’s own attempt together with the reference solution. As shown in Fig. 5, performance generally improves as the view family becomes more informative and diverse. On Qwen3-4B, moving from one to three math views improves Avg@8 by 2.9% on AIME24 and 2.8% on AIME25. Adding a fourth math view provides only a small additional gain on AIME24 and AIME25 (+0.7% and +0.2%), while remaining comparable on HMMT25, suggesting diminishing returns once the main sources of privileged information are covered.

We observe a similar trend on code, where we start from a single reference-implementation view and incrementally add complementary code views. Fig. 6 shows that moving from one to three views improves Codeforces by 2.0%. Unlike the math setting, the fourth code view continues to provide a clearer benefit, with gains of 3.3% and 0.5% over the single-view setting, respectively. Overall, these results suggest that AVSD benefits from richer teacher families across both math and code: multiple views expose more stable cross-view signal for gated reconstruction, while additional views can still help when they provide complementary information.

5 Related Work

Learning from Privileged Information. Learning from privileged information studies how training-time side information unavailable at inference can improve a model [Lopez-Paz et al., 2016]. In LLM post-training, such information can be reference solutions, final answers, demonstrations, or environment feedback. Prior work uses these signals in several ways. RLVR methods provide sparse verifiable rewards for math and code [Shao et al., 2024, DeepSeek-AI, 2025], while self-improvement and bootstrapping methods construct additional supervision from model-generated data or verified reasoning traces [Zelikman et al., 2022, Wang et al., 2023]. Other approaches use privileged information to improve exploration or curriculum construction: Qu et al. [2026] leverages oracle solution prefixes for on-policy exploration; Yan et al. [2026] learns under off-policy guidance from privileged rollouts; Chen et al. [2026a], Liao et al. [2026], Wang et al. [2026] transform privileged trajectories into abstract hints. Synthetic-curriculum methods such as Sundaram et al. [2026], Chen et al. [2026b] generate easier or more informative training instances to bridge low-signal regimes. These methods demonstrate that privileged or derived information can substantially improve learning when direct reward signal is sparse. However, they generally focus on how to

obtain better trajectories, hints, or curricula, rather than how to construct a token-level learning signal when multiple privileged views are simultaneously available. Moreover, the choice of privileged format remains consequential: different forms of privileged information vary in information density, task utility, and induced student–teacher distribution shift [Penaloza et al., 2026, Li et al., 2026b]. Our work is complementary to these approaches: rather than proposing a new source of privileged information or exploration, AVSD provides a principled way to aggregate multiple privileged views into a better learning signal by separating cross-view consensus from view-specific residual support.

On-Policy Self-Distillation and Teacher Aggregation. Self-distillation methods remove the external teacher needed in on-policy distillation by using the same model under richer conditioning. Zhao et al. [2026] condition on reference math solution, Shenfeld et al. [2026], Ye et al. [2026] distill demonstration-conditioned behavior, and reinforcement learning via self-distillation uses feedback, successful rollouts, or reward-conditioned self-revisions as privileged context [Hübotter et al., 2026, Song et al., 2026, He et al., 2026]. Penaloza et al. [2026] further study teacher-student transfer when the teacher has privileged information and the student must act without it. These methods show that the distillation objectives can serve as a useful token-level advantage. Yang et al. [2026] identify a key failure of direct privileged-teacher matching under information asymmetry and anchors the update direction to verifiable rewards while using self-distillation mainly to adjust token-level update strength. Related hybrid RL-distillation methods combine policy optimization with distillation losses [Ding, 2026, Li et al., 2026a, Xu et al., 2025], and ensemble distillation methods aggregate teachers by averaging logits or votes. However, these methods do not characterize which parts of the teacher signal are stable across views versus specific to a privileged context. In contrast, AVSD derives the aggregation structure from the reverse-KL learning signal itself. Thus, our method complements RLSD and hybrid RL-distillation: rather than relying on binary rewards (which are only available in verifiable setting) to determine the update direction for a single privileged teacher, AVSD reconstructs the token-level learning signal from the geometry of a multi-view privileged teacher family.

6 Conclusion

In this work, we introduced AVSD, a multi-view framework for privileged self-distillation. Instead of distilling from a single privileged teacher, AVSD treats different views of the same training-time artifact as a teacher family and reconstructs a better token-level learning signal from their shared structure. Experiments on math and code reasoning benchmarks show that AVSD consistently improves over single-view self-distillation and GRPO, and our analyses indicate that it produces more reliable token-level credit signals.

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A Limitations and Broader Impact

Limitations. Our work focuses primarily on reasoning tasks such as math and code. Although the formulation of AVSD is general, validating it in broader settings, such as tool-use agents, is a valuable direction for future work. The method also depends on the quality of the constructed views, as well as the underlying data quality. For example, in math reasoning, noisy annotated solutions may lead to less reliable teacher signals. In addition, AVSD is instantiated with reverse-KL on-policy self-distillation. While the reconstructed target can in principle be used with other distribution-matching losses, our advantage interpretation and bounded-residual guarantee require further study for other objectives, as discussed in Appendix B.5.

Broader Impact. This work aims to improve how language models learn from information that is available during training, such as solutions, demonstrations, or feedback. By making better use of this information, AVSD may help improve model performance on reasoning tasks such as math and code. Better models could support education, coding assistance, scientific problem solving, and other applications. However, stronger reasoning models can also be misused, and improvements in post-training methods should be accompanied by careful evaluation and deployment safeguards. In addition, training-time information may contain biases and artifacts from the data collection process. Although AVSD is designed to reduce reliance on any single privileged source, it does not remove the need for high-quality data and evaluation for unintended behavior.

B Additional Details for AVSD

In this section, we provide additional details for our method. In Appendix B.1, we derive the reverse-KL sampled-token advantage used in on-policy self-distillation. In Appendix B.2, we derive the arithmetic and geometric pooled targets and show how they correspond to natural aggregation objectives over the teacher family. In Appendix B.3, we provide further details on the gated reconstruction mechanism and the residual-boundedness property used in the main method. In Appendix B.4, we provide the detailed training algorithm for AVSD. In Appendix B.5, we discuss the applicability to other distillation losses.

B.1 Reverse-KL Token Advantage

Fix a student prefix h_t and a stop-gradient teacher distribution q_t . Let $p_t(v) = p_\theta(Y_t = v \mid h_t)$. The local reverse-KL loss is

$$\ell_t(\theta) = D_{\text{KL}}(p_t \parallel q_t) = \sum_{v \in \mathcal{V}} p_t(v) (\log p_t(v) - \log q_t(v)).$$

Taking the gradient with respect to the student parameters,

$$\begin{aligned} \nabla_\theta \ell_t &= \sum_v \nabla_\theta p_t(v) (\log p_t(v) - \log q_t(v) + 1) \\ &= \mathbb{E}_{v \sim p_t} [(\log p_t(v) - \log q_t(v) + 1) \nabla_\theta \log p_t(v)]. \end{aligned}$$

The constant 1 is a baseline because $\mathbb{E}_{v \sim p_t} [\nabla_\theta \log p_t(v)] = 0$. Therefore,

$$\nabla_\theta \ell_t = \mathbb{E}_{v \sim p_t} [(\log p_t(v) - \log q_t(v)) \nabla_\theta \log p_t(v)].$$

Equivalently, minimizing reverse KL is a policy-gradient update with distillation advantage

$$A_t(v) = \log q_t(v) - \log p_t(v),$$

since gradient descent on ℓ_t moves in the direction

$$\mathbb{E}_{v \sim p_t} [A_t(v) \nabla_\theta \log p_t(v)].$$

This is the advantage used in the main text to interpret whether a token type is promoted or suppressed. The sampled-token estimator replaces the full expectation over $v \sim p_t$ by the generated token y_t at the visited prefix.

B.2 Derivations of Two Pooled Targets

Fix a prefix h_t and write $q_m = q_t^{(m)}$.

Geometric Target. The geometric target is

$$q_t^G = \arg \min_q \frac{1}{M} \sum_{m=1}^M D_{\text{KL}}(q \parallel q_m).$$

Expanding gives

$$\frac{1}{M} \sum_m D_{\text{KL}}(q \parallel q_m) = \sum_v q(v) \log q(v) - \sum_v q(v) \frac{1}{M} \sum_m \log q_m(v).$$

Let

$$\tilde{q}_G(v) = \exp \left(\frac{1}{M} \sum_m \log q_m(v) \right).$$

The Lagrangian is

$$\mathcal{L}(q, \nu) = \sum_v q(v) \log q(v) - \sum_v q(v) \log \tilde{q}_G(v) + \nu \left(\sum_v q(v) - 1 \right).$$

Taking derivatives gives

$$\log q(v) + 1 - \log \tilde{q}_G(v) + \nu = 0,$$

so

$$q(v) \propto \tilde{q}_G(v).$$

Therefore,

$$q_t^G(v) = \frac{\tilde{q}_t^G(v)}{\sum_u \tilde{q}_t^G(u)}.$$

Arithmetic Target. The arithmetic target is

$$q_t^A = \arg \min_q \frac{1}{M} \sum_{m=1}^M D_{\text{KL}}(q_m \parallel q).$$

Expanding the terms that depend on q ,

$$\frac{1}{M} \sum_m D_{\text{KL}}(q_m \parallel q) = - \sum_v \left(\frac{1}{M} \sum_m q_m(v) \right) \log q(v) + \text{const.}$$

The minimizer of cross-entropy with target distribution

$$\frac{1}{M} \sum_m q_m$$

is exactly

$$q_t^A(v) = \frac{1}{M} \sum_m q_t^{(m)}(v).$$

The geometric and arithmetic targets also connect to classical opinion pooling: linear pools correspond to arithmetic aggregation, while log-linear pools and products of experts combine distributions through normalized products [Abbas, 2009]. The same aggregation perspective naturally extends to non-uniform view weights. If each view is assigned a prefix-level reliability weight $w_t^{(m)} \geq 0$ with $\sum_m w_t^{(m)} = 1$, then the geometric and arithmetic pooled targets become the corresponding weighted KL barycenters by replacing the uniform factor $1/M$ with $w_t^{(m)}$ in the objectives above. This yields the same interpretation: the geometric target captures weighted cross-view consensus, while the arithmetic target captures weighted view-marginal support. In our main experiments, we use uniform weights to keep the method parameter-free and avoid introducing additional view-reliability hyperparameters.

B.3 Bounded Residual Property

For the gate

$$\lambda_t(v) = C_t(v) \frac{|A_t^G(v)|}{|A_t^G(v)| + J_t(v) + \epsilon},$$

where $C_t(v) \in [0, 1]$, we have

$$0 \leq \lambda_t(v) J_t(v) = C_t(v) \frac{|A_t^G(v)| J_t(v)}{|A_t^G(v)| + J_t(v) + \epsilon} \leq |A_t^G(v)|.$$

Therefore, the reconstructed advantage

$$\hat{A}_t(v) = A_t^G(v) + \lambda_t(v) J_t(v)$$

preserves the direction of the consensus update: if $A_t^G(v) > 0$, then $\hat{A}_t(v) \geq 0$; if $A_t^G(v) < 0$, then $\hat{A}_t(v) \leq 0$. Thus, the residual can strengthen a positive consensus update or soften a negative consensus update, but cannot flip the consensus-induced direction.

B.4 Detailed Training Algorithm

Algorithm 1 summarizes the full training procedure. The method preserves the on-policy nature of self-distillation: all teacher distributions are evaluated on prefixes generated by the current student. Compared to single-view self-distillation, the only additional cost is evaluating multiple privileged views for the same prefix. In practice, these view-conditioned teacher forward passes can be batched. We report the run time comparison with single-view OPSD in Table 2. AVSD increases training time only modestly, from 41.5s to 46.7s per step, corresponding to a $1.13\times$ relative cost, since all views share the same student rollout and only add batched teacher evaluations.

Algorithm 1 Multi-View Self-Distillation with Gated Reconstruction. In our experiments, we find that $M = 3$ works well.

Require: Student model p_θ , dataset $\mathcal{D} = \{(x, r)\}$, view transformations $\{T_m\}_{m=1}^M$, small constant $\epsilon > 0$.

```

1: for each training iteration do
2:   Sample a minibatch  $\{(x_i, r_i)\} \sim \mathcal{D}$ .
3:   for each example  $(x, r)$  do
4:     Construct privileged views  $r^{(m)} = T_m(r)$  for  $m = 1, \dots, M$ .
5:     Sample an on-policy rollout  $y \sim p_\theta(\cdot | x)$ .
6:     for each prefix  $h_t = (x, y_{<t})$  do
7:       Compute student distribution  $p_t(\cdot) = p_\theta(\cdot | h_t)$ .
8:       for each view  $m = 1, \dots, M$  do
9:         Compute teacher distribution

```

$$q_t^{(m)}(\cdot) = \text{sg} \left[p_\theta(\cdot | h_t, r^{(m)}) \right].$$

```

10:      end for
11:    end for
12:    Compute

```

$$q_t^A(v) = \frac{1}{M} \sum_m q_t^{(m)}(v), \quad \tilde{q}_t^G(v) = \left(\prod_m q_t^{(m)}(v) \right)^{1/M}.$$

```

12:    Compute residual

```

$$J_t(v) = \log q_t^A(v) - \log \tilde{q}_t^G(v).$$

```

13:    Compute per-view advantages

```

$$\Delta_t^{(m)}(v) = \log q_t^{(m)}(v) - \log p_t(v), \quad A_t^G(v) = \frac{1}{M} \sum_m \Delta_t^{(m)}(v).$$

```

14:    Compute gate

```

$$C_t(v) = \frac{|A_t^G(v)|}{\frac{1}{M} \sum_m |\Delta_t^{(m)}(v)| + \epsilon}, \quad \lambda_t(v) = C_t(v) \frac{|A_t^G(v)|}{|A_t^G(v)| + J_t(v) + \epsilon}.$$

```

15:    Reconstruct teacher

```

$$q_t^*(v) = \frac{\tilde{q}_t^G(v) \exp(\lambda_t(v) J_t(v))}{\sum_{u \in \mathcal{V}} \tilde{q}_t^G(u) \exp(\lambda_t(u) J_t(u))}.$$

```

16:    Accumulate loss

```

$$\mathcal{L}_{\text{AVSD}} \leftarrow \mathcal{L}_{\text{AVSD}} + D_{\text{KL}}(p_t(\cdot) \parallel \text{sg}[q_t^*(\cdot)]).$$

```

17:  end for
18: end for
19:   Update  $\theta$  by minimizing  $\mathcal{L}_{\text{AVSD}}(\theta)$ .
20: end for

```

B.5 Applicability to other Distillation Losses

Although we derive AVSD with reverse KL, the teacher reconstruction itself is not tied to a particular divergence. The multi-view aggregation step constructs a reconstructed target q_t^* from the teacher family $\{q_t^{(m)}\}_{m=1}^M$, and this target can in principle be used with other distribution-matching losses, such as forward KL,

$$D_{\text{KL}}(q_t^* \parallel p_\theta(\cdot | h_t)),$$

or symmetric objectives such as Jensen–Shannon divergence. However, the policy-gradient interpretation in our method is specific to the reverse-KL objective: the sampled-token update is driven by

Table 2: Runtime overhead of AVSD compared with single-view self-distillation. We report average wall-clock training time per gradient update and relative overhead using Qwen3-4B under the same batch size and hardware setup. AVSD uses the same student rollouts as OPSD and only adds view-conditioned teacher evaluations.

Method	Time / step (s)	Relative time
OPSD	41.5	1.00×
AVSD	46.7	1.13×

Table 3: Training hyperparameters for GRPO and self-distillation methods.

Category	Parameter	GRPO	OPSD/AVSD
Training	Learning Rate	5×10^{-6}	5×10^{-6}
	Effective Batch Size	16	16
	Optimizer	AdamW	AdamW
	Gradient Clip Norm	0.1	0.1
	Training Steps	500	200
LoRA	LoRA Rank (r)	64	64
	LoRA Alpha (α)	128	128
	LoRA Target Modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj	
Generation	Number of Rollouts	8	1
	Sampling Temperature	1.0	0.7
	Inference Engine	vLLM	vLLM

the reconstructed advantage

$$\hat{A}_t(y_t) = A_t^G(y_t) + \lambda_t(y_t)J_t(y_t),$$

which is naturally evaluated on student-generated tokens and therefore remains on-policy. In contrast, forward KL places expectation under the reconstructed teacher q_t^* , requiring full-vocabulary supervision or teacher-sampled tokens, and does not yield the same sampled-token advantage interpretation. As a result, while the reconstructed teacher q_t^* can be paired with other distillation losses, our main design and guarantees are most directly aligned with reverse-KL on-policy self-distillation. Exploring whether the same multi-view reconstruction improves forward-KL or symmetric distillation objectives is an interesting direction for future work.

C Implementation Details of Experiments

Training and Evaluation Configuration. We provide the detailed training hyperparameters for different methods in Table 3. Following prior work [Zhao et al., 2026, Li et al., 2026a], we disable the Qwen3 thinking mode during training. Prior work has also observed that Qwen models with thinking mode enabled tend to produce long, exploratory chains of thought, which can be harder to learn from [Kim et al., 2026]. Therefore, we leave this as an open problem for future work.

For evaluation, we use vLLM [Kwon et al., 2023] with the following settings for all models: temperature = 0.6, top- p = 0.95, top- k = 20, min- p = 0, and max tokens 38912.

GPUs. Most of the experiments are run on 4 A6000 GPUs. Some experiments such as training for Qwen3-8B and GRPO baselines are run on 4 A100 GPUs.

D Additional Experiments

Effect of Privileged-View Choice. In Fig. 1, we show that for math reasoning no single privileged view is uniformly optimal across benchmarks. We further examine whether the same phenomenon

holds in the code domain on Qwen3-8B. As shown in Table 4, different code views lead to different strengths: the reference-implementation view performs best among single-view OPSD variants on Codeforces, while the feedback view performs best on LiveCodeBench and achieves the strongest single-view average. This confirms that the optimal privileged view is task-dependent in code as well. In contrast, AVSD combines all views and achieves the best performance on both benchmarks, improving over the strongest single-view OPSD baseline by 2.6% on Codeforces and 1.5% on LiveCodeBench. These results further support our motivation that multi-view self-distillation can construct a stronger and more robust learning signal than committing to any single privileged view.

Table 4: Effect of privileged-view choice in code self-distillation. We report Avg@8 on Codeforces and LiveCodeBench with Qwen3-8B.

Method	View(s)	Codeforces	LiveCodeBench	Avg.
Base		60.9	47.6	54.3
OPSD	code	63.2	50.3	56.8
OPSD	hint	61.3	48.2	54.8
OPSD	feedback	62.8	50.9	56.9
AVSD (Ours)	all	65.8	52.4	59.1

Gate Analysis. We analyze how often AVSD uses the residual pathway during training. Specifically, we report the percentage of token positions for which the gate is open, indicating that the method adds residual support from the arithmetic target beyond the geometric consensus signal. As shown in Table 5, the gate-open rate varies across models but remains consistently between 21% and 27%. This suggests that AVSD does not simply collapse to either the consensus or arithmetic target: it primarily relies on stable cross-view consensus, while selectively incorporating residual information for tokens where the multi-view signal is reliable.

Token-Level Credit Allocation. We provide supplementary token-level credit results for the main paper in Fig. 7 and Fig. 8. Fig. 7 expands the qualitative example from Fig. 4 by showing the full problem, the generated trace, and the token credits assigned by each method, showing the benefits of AVSD over baselines. Fig. 8 quantifies the reliability of the learning signal: GRPO provides no token-level signal in 29.6% of incorrect-rollout cases, OPSD assigns incorrect-sign credit to 12.3% of high-impact tokens, while AVSD reduces this rate to 2.9%.

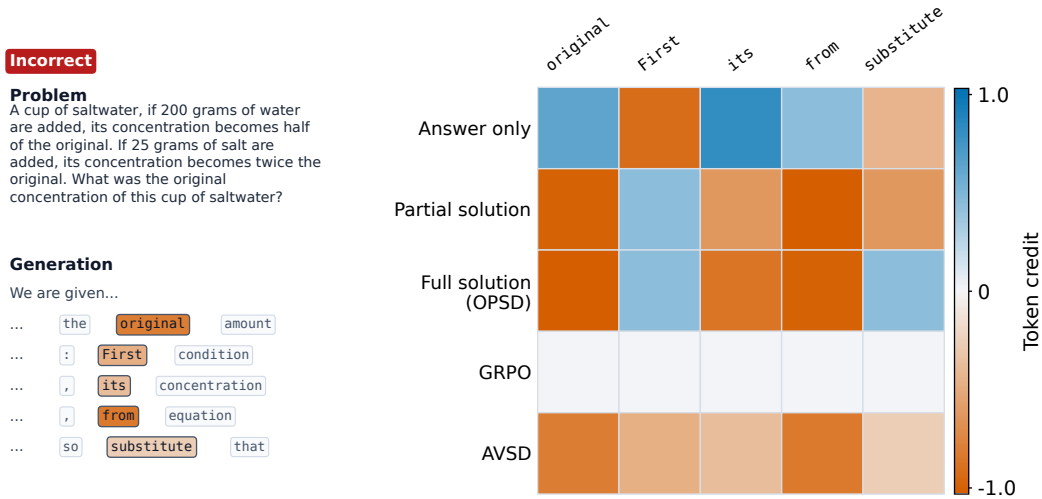


Figure 7: The full problem with generation trace for Fig. 4.

Model	Gate Open (%)
Qwen3-4B	24.6
Qwen3-8B	21.1
DeepSeek-R1-Distill-Qwen-7B	26.6

Table 5: Gate-open rate during training. We report the percentage of token positions for which the residual gate is open.

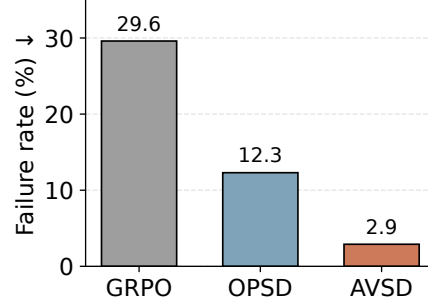


Figure 8: Quantitative reliability of token-level learning signals. For OPSD and AVSD, we report the percentage of top generated tokens with the largest absolute credit magnitude whose assigned credit has the incorrect sign. For GRPO, we report the percentage of cases with no learning signal, which occurs when no sampled rollout is correct. Lower is better.

E Examples of Multi-View Privileged Information

Example of Multi-View Privileged Information in Math

Question. Real numbers a, b, c, d are such that $a + b = \frac{9}{c-d}$ and $c + d = \frac{25}{a-b}$. What is the minimum value that the quantity $a^2 + b^2 + c^2 + d^2$ can take?

Full Solution View Given the equations involving real numbers a, b, c , and d :

$$a + b = \frac{9}{c-d}$$

$$c + d = \frac{25}{a-b}$$

We aim to find the minimum value of $a^2 + b^2 + c^2 + d^2$.

1. ****Multiply both given equations by their respective denominators****:

$$(a + b)(c - d) = 9$$

$$(c + d)(a - b) = 25$$

2. ****Expand and rearrange the products****:

$$ac - ad + bc - bd = 9$$

$$ac - bc + ad - bd = 25$$

3. ****Add the two equations to eliminate terms****:

$$2(ac - bd) = 34 \implies ac - bd = 17$$

4. ****Establish a lower bound for $a^2 + b^2 + c^2 + d^2$ ****:

Since $a^2 + c^2 \geq 2ac$ (which follows from the AM-GM inequality $(a - c)^2 \geq 0$) and $b^2 + d^2 \geq -2bd$ (which follows from $(b + d)^2 \geq 0$):

$$a^2 + c^2 \geq 2ac$$

$$b^2 + d^2 \geq -2bd$$

Adding these inequalities yields:

$$a^2 + b^2 + c^2 + d^2 \geq 2(ac - bd)$$

Given $ac - bd = 17$:

$$a^2 + b^2 + c^2 + d^2 \geq 2 \times 17 = 34$$

5. ****Check if the equality can be achieved****:

Equality in the inequalities $a^2 + c^2 \geq 2ac$ and $b^2 + d^2 \geq -2bd$ is achieved when $a = c$ and $b = -d$.

Substitute $a = c$ and $b = -d$ into the original equations. The first equation becomes:

$$a + b = \frac{9}{c - d} \implies a - d = \frac{9}{a - d} \implies (a - d)^2 = 9.$$

The second equation becomes:

$$c + d = \frac{25}{a - b} \implies a + d = \frac{25}{a + d} \implies (a + d)^2 = 25.$$

Taking positive roots gives $a - d = 3$ and $a + d = 5$, so $a = 4$ and $d = 1$. Hence $c = a = 4$ and $b = -d = -1$. Substituting back:

$$a^2 + b^2 + c^2 + d^2 = 4^2 + (-1)^2 + 4^2 + 1^2 = 16 + 1 + 16 + 1 = 34.$$

This confirms the minimum value is achievable.

Conclusion

$$\boxed{34}$$

Partial Solution View Given the equations involving real numbers a, b, c , and d :

$$a + b = \frac{9}{c - d}$$

$$c + d = \frac{25}{a - b}$$

We aim to find the minimum value of $a^2 + b^2 + c^2 + d^2$.

1. ****Multiply both given equations by their respective denominators****:

$$(a + b)(c - d) = 9$$

$$(c + d)(a - b) = 25$$

2. ****Expand and rearrange the products****:

$$ac - ad + bc - bd = 9$$

$$ac - bc + ad - bd = 25$$

Final Answer View

$$\boxed{34}$$

Example of Multi-View Privileged Information in Math

Question. In a company, employees have a combined monthly salary of \$10,000. A kind manager proposes to triple the salaries of those earning up to \$500, and to increase the salaries of others by \$1,000, resulting in a total salary of \$24,000. A strict manager proposes to

reduce the salaries of those earning more than \$500 to \$500, while leaving others' salaries unchanged. What will the total salary be in this case?

Full Solution View

1. Let us denote by x the total number of employees whose monthly salary is less than or equal to 500 dollars.
2. Let y be the total number of employees whose monthly salary is more than 500 dollars.
3. The firm's total monthly salary expenditure before any changes is 10,000 dollars. This can be written as:

$$500x + \sum_{i=1}^y s_i = 10,000$$

where s_i represents the individual salaries of the employees earning more than 500 dollars.

4. The kind manager proposes the following changes: - Tripling the salaries of employees earning 500 dollars or less. - Increasing the salaries of employees earning more than 500 dollars by 1000 dollars each.

The new total salary expenditure proposed by the kind manager is 24,000 dollars. This can be expressed as:

$$3 \cdot 500x + \sum_{i=1}^y (s_i + 1000) = 24000$$

5. Simplifying the equation:

$$1500x + \sum_{i=1}^y s_i + 1000y = 24000$$

6. Since we know that $\sum_{i=1}^y s_i = 10000 - 500x$ from the original total monthly salary equation, substituting this we get:

$$1500x + (10000 - 500x) + 1000y = 24000$$

7. Simplifying further, we obtain:

$$1000x + 1000y = 14000 \implies x + y = 14$$

8. The cruel manager proposes the following changes: - Reducing the salaries of employees earning more than 500 dollars to 500 dollars each. - Keeping the salaries of employees earning 500 dollars or less unchanged.

The new total salary can be calculated as:

$$500x + 500y = 500(x + y)$$

9. From step 7, we know that $x + y = 14$, so substituting this into the equation for the total salary proposed by the cruel manager:

$$500 \cdot 14 = 7000$$

10. Conclusion: The new total salary expenditure after the cruel manager's proposed changes will be:

7000

Partial Solution View

1. Let us denote by x the total number of employees whose monthly salary is less than or equal to 500 dollars.
2. Let y be the total number of employees whose monthly salary is more than 500 dollars.
3. The firm's total monthly salary expenditure before any changes is 10,000 dollars. This can be written as:

$$500x + \sum_{i=1}^y s_i = 10,000$$

where s_i represents the individual salaries of the employees earning more than 500 dollars.

Final Answer View

7000

Example of Multi-View Privileged Information in Code

Question.

Given n sticks with lengths a_i , find the maximum number of regular equal-sided polygons you can construct. Each side uses one stick, and no stick can be reused.

Reference Solution View

```
import sys
from collections import Counter

t = int(sys.stdin.readline())
for _ in range(t):
    n = int(sys.stdin.readline())
    a = list(map(int, sys.stdin.readline().split()))
    cnt = Counter(a)
    total = 0
    for v in cnt.values():
        total += v // 3
    print(total)
```

Hint View

Greedy form triangles from sticks of the same length. For S sticks of one length, you can create $\lfloor S/3 \rfloor$ polygons. Sum this over all stick lengths.

Feedback View

Wrong student rollout:

```
import sys
input=sys.stdin.readline
t=int(input())
for _ in range(t):
    n=int(input())
    a=list(map(int,input().split()))
    print(0)
```

Feedback: The student's solution failed on a test case.

Failing input:

```
4
1
1
2
1 1
6
2 2 3 3 3 3
9
4 2 2 2 2 4 2 4 4
```

Expected output:

```
0
0
1
2
```

Student output:

```
0
0
0
0
```

Example of Multi-View Privileged Information in Code

Question.

$S(x)$ is the sum of digits of x . Count integers x with $1 \leq x \leq n$ such that

$$S(x+1) < S(x).$$

Reference Solution View.

```
t = int(input())
for _ in range(t):
    n = int(input())
    print((n + 1) // 10)
```

Hint View.

A number is interesting exactly when it ends in 9. Count numbers $\leq n$ whose last digit is 9, which is $\lfloor (n+1)/10 \rfloor$.

Feedback View.

Wrong student rollout:

```
t=int(input())
for _ in range(t):
    n=int(input())
    print(n//10)
```

Feedback: The student's solution failed on a test case.

Failing input:

```
5
1
9
10
34
880055535
```

Expected output:

```
0
1
1
3
88005553
```

Student output:

```
0
0
1
3
88005553
```

F Prompts

Teacher Prompt

Problem: {problem}

Here is training-time reference information for this problem. It may contain a final answer, a hint, a partial solution, or a full solution to the problem.

=== Privileged Reference ({view_type}) Begin ===


```
{privileged information}  
=== Privileged Reference ({view_type}) End ===  
Use the reference only as hidden guidance. Use it only to understand,  
verify, or correct your solution.  
Do not mention, quote, copy, cite, or refer to the privileged  
reference or its existence.  
Now solve the original problem independently using your own  
reasoning.  
Please reason step by step, and put your final answer within  
\boxed{}
```

Student Prompt

```
Problem: {problem}  
Please reason step by step, and put your final answer within  
\boxed{}
```